

THE OECD TAX-BENEFIT MODEL FOR THE UNITED STATES

Description of policy rules for 2018



THE OECD TAX-BENEFIT MODEL FOR THE UNITED STATES

Description of policy rules for 2018

OECD team: Marion Bachelet, James Browne, Herwig Immervoll, Dirk Neumann (lead author), Daniele Pacifico, Olga Rastrigina

National team: Department of Health and Human Services: Donald Oellerich (lead), Nina Chien (reference person), Abdihakim Abdi, Peter Germanis, Susan Hauan; Department of Labor: Daniel Hayes

This version: December 2018 – via <http://www.oecd.org/els/benefits-and-wages.htm>

REFERENCE: No. VS/2015/0427 (DI150030 & DI151310)

This work is published on the responsibility of the Director of the OECD Directorate for Employment, Labour and Social Affairs. The opinions expressed and arguments employed herein do not necessarily reflect the official views of the Organisation or of the governments of its member countries.

This document and any map included herein are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

© OECD 2018

You can copy, download or print OECD content for your own use, and you can include excerpts from OECD publications, databases and multimedia products in your own documents, presentations, blogs, websites and teaching materials, provided that suitable acknowledgment of the source and copyright owner is given. All requests for public or commercial use and translation rights should be submitted to rights@oecd.org. Requests for permission to photocopy portions of this material for public or commercial use shall be addressed directly to the Copyright Clearance Center (CCC) at info@copyright.com or the Centre français d'exploitation du droit de copie (CFC) at contact@cfcopies.com.

Table of contents

<i>Preface</i>	4
<i>The OECD tax-benefit model for the United States: Policy rules in 2018</i>	5
1. Reference wages	5
2. Unemployment benefits	5
2.1. Unemployment insurance	5
2.2. Unemployment assistance	7
3. Social assistance and housing benefits	7
3.1. Supplemental Nutrition Assistance Program (SNAP)	7
3.2. Housing benefits	9
4. Family benefits	10
4.1. Temporary Assistance for Needy Families (TANF)	10
5. Childcare for pre-school children	11
5.1. Gross childcare fees	12
5.2. Fee discounts and free provision (Head Start Program)	12
5.3. Childcare benefit for formal centre-based care (Child Care and Development Fund, CCDF)	13
5.4. Childcare allowance for children not using childcare centres	15
5.5. Tax concessions for childcare expenditures (Child and Dependent Care Credit, CDCC)	15
6. In-work benefits	15
6.1. Earned Income Tax Credit (EITC)	15
7. Social security contributions and payroll taxes	16
7.1. Employee's social security contributions	16
7.2. Employer's social security contributions	16
7.3. Payroll taxes	16
8. Taxes	17
8.1. Personal income tax	17
8.2. Local income taxes	18
<i>Annex: Other benefits and direct taxes</i>	19
Supplemental Security Income (SSI)	19
General Assistance programme (GA)	19
Housing benefits	19

Preface

The [OECD Tax-Benefit model \(TaxBEN\)](#) incorporates detailed policy rules for tax liabilities and benefit entitlements as they apply to individual families across OECD member countries. Its main use is to calculate the amount of taxes that people are liable to pay, and the government transfers they are likely to receive, in different family and labour-market situations. The model includes legal policy rules that are relevant for people of **working age** (from 18 years old until the statutory retirement age) and their dependent children. Income tax liabilities and benefit entitlements are calculated for a broad set of *stylised* families (“vignettes”, e.g. a married couple of 40 years old adults with two children aged 4 and 6 respectively). Model users are free to change many of these characteristics, including the age and number of children, activity status of adult members, hours of work, current and past earnings levels, unemployment duration, social contribution records, and housing-related costs. The model has been updated annually since the early 2000s for most OECD countries.

TaxBEN’s policy scope includes the main taxes on employment income (earnings), social contributions paid by individuals and by employers, as well as the main cash and near-cash benefit programmes, including unemployment benefits, family benefits, guaranteed minimum-income benefits, cash housing benefits, and employment-conditional benefits. Disability benefits and support for non-parental childcare are included for a sub-set of countries and years. The most important policy areas that are outside the scope of the model include taxes on wealth (e.g. taxes on immovable and unmovable properties, including local taxes), indirect taxes (e.g. VAT), early-retirement benefits, sickness benefits and in-kind transfers (e.g. free school meals, subsidised transport and free health care).

This report describes the taxes and benefits that are included in the model and focuses on the rules that are relevant for family, individual and labour-market circumstances that are within its scope. The **Annex** provides information on other cash benefits and taxes on employment incomes that can be relevant for some members of the working-age population, but which are not included in the TaxBEN model.

Reading notes and further details on the scope and content of this report

- The **reference date** for policy rules described in this report is **January 1, 2018**.
- **Guidelines for completing and updating this report** are provided [here](#).
- Further information on the model, model results, and references to reports and analytical uses is available on the [project website](#). A [methodology](#) document provides a full description of the assumptions underlying the model as well as the model choices that users can make. The symbol  in the text provides a link to a glossary of technical terms.
- Section titles provide the names of taxes and benefits as they are known in the country: first, direct translation into English, then (in brackets) the name in the national language.
- In order to facilitate transparency between the policy descriptions and the associated code in the model, the variable names are indicated in the text in square brackets using the following format: **[variable name]**, for instance: **[AW]** for the average wage.

The OECD tax-benefit model for the United States: Policy rules in 2018

1. Reference wages

Average wage: The 2018 average full-time wage **[AW]** was USD 54,396 (preliminary Secretariat estimate).¹

The federal minimum wage **[MIN]** in 2018 is USD 7.25 per hour. The annual minimum wage in the OECD tax-benefit model is computed by multiplying the minimum hourly wage (as of January 1, 2018) by 40*52, i.e. USD 7.25*40*52 = USD 15,080.

2. Unemployment benefits

2.1. Unemployment insurance

Variable name:² **[UB; UI_p; UI_s]**

This benefit is contributory, not means-tested and taxable. 

2.1.1. Eligibility conditions

Each State administers its own programme. Eligibility conditions differ from State to State, as do maximum benefit levels. This section describes the system in Michigan, which is modelled in TaxBEN.

Earnings requirement: To qualify for the minimum weekly benefit amount (WBA), High Quarter Wages (HQP) must equal USD 3,589 and Base Period Wages (BPW) must equal USD 5,384 with BPW being 1½ times the HQP, with wages in 2 quarters. Alternatively, BPW must equal 20 times the state average weekly wage (SAWW) with wages in 2 quarters (USD 19,826 in BPW).³

Minimum employment record: Whether an employing unit is an employer depends on the number of days or weeks a worker is employed or the amount of the employing unit's quarterly or yearly payroll. Except for agricultural labor and domestic service, if an employing unit or employer paid wages of \$1,500 or more in any calendar quarter in the

¹ AW refers to the Average Wage estimated by the [Centre for Tax Policy and Administration](#). For more information on methodology see the latest [Taxing Wages publication](#). The AW calculations for the US assume that the average worker lives in Detroit, Michigan.

² The variable names ending with “_p” refer to the first adult (so-called “principal” adult) whereas those ending with “_s” are related to the spouse.

³ The base period comprises 4 quarters (according to the “standard” or “alternate” base period). The HQP is the quarter with the highest wages out of these 4 quarters.

current or immediately preceding calendar year, or if the employing unit had one or more workers on at least one day in each of 20 weeks during the current or immediately preceding calendar year.

Behavioural requirements:⁴

Generally, all States require that UI recipients be able to work and available for full-time work. All states also require that UI recipients be actively seeking work. TaxBEN assumes “involuntary” job loss and that the behavioural requirements are satisfied.

2.1.2. Benefit amount

The weekly benefit amount (WBA) in Michigan is calculated as the 4.1 percent of HQW during the Base Period plus USD 6 for each dependent up to 5 dependents, maximum WBA is the same with or without dependents. The benefit is bound by a minimum WBA of USD 147 (USD 147*14 weeks (see Section 2.1.3) = USD 2,058 per benefit year), and a maximum of USD 362 per week (USD 362*20 weeks (see Section 2.1.3) = USD 7,240 per benefit year).

2.1.3. Benefit duration

Workers get regular state UI benefits for 14 to 20 weeks, based upon amount of wages earned. No waiting period is required. If they are still unemployed after those benefits are exhausted, they will draw Extended Benefits if the State has triggered on Extended Benefits and is in an Extended Benefits period.⁵

Maximum duration of benefits in Michigan on January 1, 2018 was 20 weeks.

2.1.4. Means test

See Section 2.1.7.

2.1.5. Tax treatment

Unemployment insurance benefit income is subject to both Federal and State government income tax, but is exempted from social security taxes.

2.1.6. Interaction with other components of the tax-benefit system

None.

⁴ Details on behavioural requirements and sanction provisions for unemployment benefits are reported in regularly updated companion reports, see Immervoll and Knotz (2018, forthcoming), [Langenbucher \(2015\)](#) and [Venn \(2011\)](#).

⁵ States that qualify for the Extended Benefit (EB) are those with a 13-week IUR (insured unemployment rate) of 5 percent or higher, and 120 percent above the IUR rate for the corresponding 13-week period in the two previous years. Michigan qualified for EB between January 25, 2009 and February 18, 2012. On January 1, 2018, EB was not payable. The maximum number of weeks of EB is 13 or 50 percent of the total amount of regular state UI benefits, whichever is less. The amount is the same as under the regular State UI benefit.

2.1.7. Combining benefit receipt and employment/starting a new job

The benefit is compatible with work but earnings reduce the amount of unemployment benefits. The weekly benefit amount (WBA) is reduced by 50 cents for every dollar earned during that week. Individuals earning more than 1.5 times their WBA are ineligible to receive benefits for that week.

2.2. Unemployment assistance

None.

3. Social assistance and housing benefits

3.1. Supplemental Nutrition Assistance Program (SNAP)

Variable name: **[SA]**

This is a non-contributory benefit, means-tested and not taxable.

SNAP is in place since 2008 and was formerly known as the *Food Stamp Program*. It is designed primarily to increase the food purchasing power of eligible low-income households to a point where they can buy a nutritionally adequate low-cost diet. This benefit scheme is classified as ‘social assistance’ in TaxBEN. The Temporary Assistance for Needy Families (TANF) is covered in Section 4. Other social assistance programmes such as State General Assistance (GA) and Supplemental Security Income (SSI) are beyond the scope of TaxBEN but are briefly described in the Annex.

Since 2002 the participation (“take-up”) rate for Food Stamps/SNAP had been rising, and 2014 was the first year that the rate declined (88.1 percent). In 2015 the rate rose slightly to 88.8 percent (most recent data available).⁶

3.1.1. Eligibility conditions

Households who meet the income and asset tests described below and who meet other requirements (such as work registration and immigration rules, for example) are eligible for SNAP benefits. To be entitled to the benefit, households need to pass two income tests and an asset test except for households where all members receive TANF, GA or SSI, who qualify automatically⁷:

⁶ See <https://aspe.hhs.gov/system/files/pdf/259196/WELFAREINDICATORS17THREPORT.pdf>, p. 22.

⁷ This automatic qualification is referred to as “*categorical eligibility*”. Households in which all members are receiving benefits through TANF, SSI, or GA are categorically eligible for SNAP. In addition, States are required to confer categorical eligibility on households receiving benefits or services that are at least 50 percent funded by TANF and have the option to confer categorical eligibility on households receiving benefits or services that are less than 50 percent funded by TANF. Many States have chosen to provide a TANF-funded noncash benefit to confer categorical eligibility on a broad number of households; this is known as broad-based categorical eligibility. As of August 2016, 42 States, including Michigan, had adopted broad-based categorical eligibility policies. Michigan applied broad-based categorical eligibility to any household with gross income less than or equal to 200 percent of poverty and with total assets less than or equal to 200 percent

- Basic (gross) monthly income must not exceed 130 percent of the poverty guidelines;
- Counted (net) monthly income must not exceed 100 percent of the poverty guideline;
- Countable assets cannot exceed USD 3500 for households with an elderly or disabled member, USD 2250 for other households (not considered in TaxBEN).

The net income guideline for a family of four in 2018 was USD 2050 per month (see table below for further family types).

Household with elderly (aged 60 and over) members do not need to meet the basic (gross) income guideline and are waived from other programme requirements.

3.1.2. *Benefit amount*

Basic (gross) monthly income is the cash household income. Earned income before federal, state and local taxes, and social security contributions is counted, as is TANF, Social Security income, Supplemental Security Income benefits, child support payments, retirement pensions, and virtually all other unearned income. The Earned Income Tax Credit (EITC) (see Section 6) is not included in basic monthly income. Also excepted is unanticipated, irregular or infrequent income up to USD 30/quarter and income from tax refunds.

In Fiscal Year (FY) 2018, net monthly income was calculated by deducting the following from gross monthly income:

- a standard deduction for all SNAP households: USD 160 for households of 3 persons or less; USD 170 for 4-person households; USD 199 for 5-person households and USD 228 for 6 or more persons⁸;
- 20 per cent of gross earnings in recognition of taxes and work related expenses;
- Court ordered child support payments made to non-household members (not modelled in TaxBEN);
- Out-of-pocket medical expenses for elderly or disabled household members in excess of USD 35 as well as out-of-pocket dependent-care expenses necessary for work, school, or while seeking employment (not modelled in TaxBEN);
- Excess shelter expense (rent or mortgage and utility expenses) if shelter expenses exceed 50 per cent of gross income less the above deductions, with a maximum of USD 535 per month for households with no elderly or disabled members.

As low-income families are expected to spend 30 per cent of their net income on food, the maximum benefit amounts are decreased by 30 per cent of counted (net) income.

A minimum monthly payment of USD 15 per month is applicable to one or two person households entitled to SNAP. Maximum Monthly SNAP allotments are linked to family size (see table).

of poverty and with total assets less than or equal to USD 5,000 after excluding the value of one vehicle and the combined value of any remaining vehicles up to USD 15,000.

⁸ Beginning in FY 2003 the standard deduction is equal to 8.31 percent of the poverty guideline which varies by household size (see in the table below).

Maximum monthly SNAP allotments for FY 2017-2018 (in USD per month)			
Household size (persons)	Maximum allotments	Gross income eligibility limit	Net income eligibility limit (Poverty Guideline)
1	192	1307	1005
2	352	1760	1354
3	504	2213	1702
4	640	2665	2050
5	760	3118	2399
6	913	3571	2747
7	1009	4024	3095
8	1153	4477	3444
Each additional person	+144	+453	+349

3.1.3. *Benefit duration*

No time limit, as long as eligibility conditions are met for most participants. SNAP benefits are issued monthly. Households are required to periodically submit information in order to be recertified for SNAP. Healthy, childless adults aged 18-49 are subject to strict work requirements and may receive benefits for only three months in any 36-month period unless they work, meet work requirements, are exempted under other provisions of law, or live in an area waived from work requirements due to insufficient jobs. In 2018, Michigan had a state-wide waiver due to employment conditions.

3.1.4. *Means test*

See Sections 3.1.1, 3.1.2 and 3.1.7.

3.1.5. *Tax treatment*

SNAP benefits are not taxable.

3.1.6. *Interaction with other components of the tax-benefit system*

SNAP benefits are not included in the means test of any other benefit.

3.1.7. *Combining benefit receipt and employment/starting a new job*

20 per cent of gross earnings are disregarded in SNAP income test in recognition of taxes and work related expense.

3.2. *Housing benefits*

The three major federal rental assistance programs are: 1) public housing; 2) project-based Section 8; 3) tenant-based Section 8 vouchers. Individual states are permitted to use funds from a limited number of other federal programmes to provide rental housing assistance. For example, some states choose to allocate a portion of funding from the Temporary Assistance to Needy Families (TANF) programme to operate rental subsidy programs, though these programs generally offer assistance that is short in duration and for limited numbers of specific target populations. Given long waiting periods for some of these benefits and the level of discretion in their admission, they are not covered in TaxBEN. Some further details can be found in the Annex.

4. Family benefits

4.1. *Temporary Assistance for Needy Families (TANF)*

Variable name: **[FB]**

TANF provides families with children with financial assistance and related support services. In Michigan this benefit is called Family Independence Program (FIP).

This is a non-contributory benefit, means-tested and not taxable.

Note however that actual participation (“take-up”) rates for TANF have been declining continuously since the program’s creation. A historically low 26.3 percent of poor eligible children and families actually received TANF benefits in 2015 (most recent data available).⁹

4.1.1. *Eligibility conditions*

Although States may impose various conditions on the receipt of assistance, a family must, based on Federal law, include at least one minor child (in Michigan, 18 years or younger, 19 years if student) or the mother must be pregnant. Also, the applicant must be a US national, citizen, or permanent resident.

4.1.2. *Benefit amount*

There are no Federal TANF rules or requirements regarding the State’s calculation of benefits. Each State may establish its own benefit levels and determine its own benefit calculation. The benefit is calculated based on the number of family members using the following monthly amounts in January 2018 (Michigan; TANF benefit levels have not changed since 2009 in Michigan):

- 1 person: USD 306 (e.g. pregnant lone mother expecting first child)
- 2 people: USD 403
- 3 people: USD 492
- 4 people: USD 597
- 5 people: USD 694
- 6 people: USD 828
- 7 people: USD 905

4.1.3. *Benefit duration*

Eligibility and benefits are determined monthly. Federal funding for a family with an adult receiving federally funded TANF assistance is limited to 60 months. The 60 months do not have to be consecutive, but it is a lifetime limit. States may use their own funds to provide benefits after the expiration of the 60 months or exclude families from the accumulation of months toward the federal 60-month limit. States may also create their own time limits (in addition to the federal time limit). Many States have either shortened

⁹ See <https://aspe.hhs.gov/system/files/pdf/259196/WELFAREINDICATORS17THREPORT.pdf>, p. 21.

the time limit (for example, several States have a 24 month time limit) or limited the number of months that a family may receive benefits within a certain period of time. For example, the family may receive benefits for 24 months within a 60 month period, but there is a lifetime limit of 60 months of federally-funded assistance with up to 20 percent of the caseload exempted from the Federal time limit due to hardship.

Michigan did not enforce a time limit from 1996 to 2007 but has enacted a 48-month time limit as of October 2007, regardless of the funding source (federal or state funds). The time limit includes exemptions for age (65 or older), a verified disability or long-term incapacity, and when a spouse or parent provides care for a disabled family member living in the home.

4.1.4. *Means test*

TANF benefits are means-tested against income. Except where another Federal statute specifies that certain income or other benefits should be disregarded, each State may decide which income to consider in calculating the benefit amount. See Section 4.1.7 for the treatment of earned income.

4.1.5. *Tax treatment*

Family benefits are not taxable.

4.1.6. *Interaction with other components of the tax-benefit system*

TANF benefits are included in the means-test for SNAP, see Section 3.1.

4.1.7. *Combining benefit receipt and employment/starting a new job*

Although this benefit is compatible with earnings there is considerable variation among the States in their treatment of earned income. In calculating the monthly benefit, Michigan disregards the first USD 200 of income earned plus 20 per cent of any additional earnings to determine initial eligibility; once eligibility has been established, it disregards the first USD 200 of income earned plus 50 per cent of any additional earnings.

Across all States, the fixed income disregard ranges from zero to USD 250 and the variable disregard from zero to 100 per cent up to a limit, which can be based on income and/or time since starting work. In general, States with high fixed disregards tend to have smaller variable disregards and vice versa. Some States (not Michigan) impose a "family cap" on benefits. The initial benefit is based on the size of the family at the time of application, and benefits do not increase for additional children conceived after eligibility is determined.

5. Childcare for pre-school children

The **reference date** for the policy rules described in this section is **January 1, 2018**.¹⁰

¹⁰ The childcare module of TaxBEN is updated every three years (next update: 2021).

5.1. Gross childcare fees

Variable name: `[UScc_cost]`

Child care services are primarily provided through a market-based system at rates determined by market forces in the priced child care market. Rates that providers charge to parents vary substantially based on region, state, age of child, and type of child care setting. In 2017 (most recent data available), the average annual cost of full-time, centre-based child care in Michigan was USD 10,603 for infants and USD 8,678 for a 4-year old.¹¹

5.1.1. Discounts for part-time usage

Michigan pays its child care subsidies based on an hourly rate, which would be the same for full-day and part-time care. See Section 5.3.2.

5.2. Fee discounts and free provision (Head Start Program)

With respect to expenses for dependent care, a household is entitled to a tax credit for a portion of the actual cost of the care of a dependent if the care enables a household member to accept or continue employment, or training or education that is preparatory for employment (Child and Dependent Care Credit, CDCC). The tax credit is non-refundable, so families that do not pay taxes do not benefit from the credit. See Section 5.5 for details.

The federal Head Start Program¹² provides free centre-based care to children aged 3 and 4 from low-income families (a very limited number of slots are also available to infants aged 0 through 2 via the Early Head Start Program). Family incomes (before taxes) must be below the federal poverty line, which in FY 2017-18 for a family of 3 was USD 20,420 (see also Section 3.1 for Poverty Guidelines). Children in foster care, homeless children, and children from families receiving public assistance (TANF or SSI) are eligible regardless of income. Additionally, programs are allowed to enroll a limited number of children (up to 35 percent of total enrollment) with incomes between the federal poverty line and 135 percent of the federal poverty line. Families do not pay a fee. TaxBEN models the Head Start Program for children aged 3 and 4 for families with income below the federal poverty line.

Many states also have preschool programs that serve 3 and 4-year olds, though these programs vary dramatically across states in terms of numbers served, eligibility requirements, and parent co-payment and are therefore not considered in TaxBEN. Across states in 2017 (most recent data available), an (unweighted) average of 33 percent of 4-year olds are served by state preschool, and an (unweighted) average of 5 percent of 3-year olds are served by state preschool. Average state spending per child is USD 5,008.¹³ In Michigan in 2017, 33 percent of 4-year-olds were served and no 3-year-olds were served. The income eligibility threshold was 250 percent of the federal poverty line (USD

¹¹ <http://usa.childcareaware.org/advocacy-public-policy/resources/research/statefactsheets/>

¹² <https://www.benefits.gov/benefits/benefit-details/1919>

¹³ http://nieer.org/wp-content/uploads/2018/04/YB2017_Executive-Summary.pdf

51,950 for a family of three). The cost per child, paid by the government, was USD 6,356.¹⁴

5.3. *Childcare benefit for formal centre-based care (Child Care and Development Fund, CCDF)*

Variable name: `[cc_subsidy/ cc_benefit]`

The Child Care and Development Fund (CCDF) is a multibillion-dollar Federal and State partnership programme. As a block grant, CCDF provides funding to States, territories, and tribes to provide child care subsidies through vouchers or certificates to low-income families, and grants and contracts with providers in some States. CCDF provides access to child care services for low-income families so parents can work, attend school, or enrol in training. Additionally, CCDF promotes the healthy development of children by improving the quality of early learning and afterschool experiences. In FY 2017, CCDF provided 5.8 billion in discretionary and mandatory matching funds to 56 States and territories and 260 tribal grantees encompassing over 500 federally recognized tribes. Each month, CCDF serves approximately 1.4 million children under the age of 13 from 850,000 low-income working families.¹⁵ Each State has discretion in determining how its CCDF child care programmes will operate. CCDF subsidies may be used to pay for care offered in non-center, family settings. TaxBEN uses CCDF as implemented in Michigan.

5.3.1. *Eligibility*

Parents must be working, or seeking employment, or attending job training or education. Typically a child must be under age 13, however, children with special needs may be eligible up to 19 years of age.¹⁶ According to the most recent CCDF Plans (FY 2017-2018) the eligibility thresholds based on income for a family of three ranged from USD 1,423 per month in Puerto Rico to USD 5,040 per month in Vermont Delaware. In 2018, Michigan's eligibility threshold for a family of three is USD 2,213 per month.¹⁷

The Federal law requires States to provide a minimum of 12 months of eligibility to children of families receiving CCDF assistance, regardless of increases in parents' earnings as long as their income remains at or below the eligibility threshold of 85 percent of state medium income (SMI).¹⁸

¹⁴ http://nieer.org/wp-content/uploads/2018/04/Michigan_YB2017.pdf

¹⁵ <https://www.acf.hhs.gov/occ/resource/ccdf-statistics>, Table 1 - Average Monthly Adjusted Number of Families and Children Served. Note however that this is only a small fraction of children eligible for child care subsidies. More details on eligibility versus actual participation ("take-up") rates can be found here (for FY 2013): <https://aspe.hhs.gov/system/files/pdf/258491/ChildCareSubsidyEligibility.pdf>

¹⁶ As TaxBEN focusses on pre-school children, the age limit applied in the model is below 6 years.

¹⁷ <https://dhhs.michigan.gov/OLMWEB/EX/RF/Public/RFT/270.pdf>

These thresholds reflect the income level needed to initially qualify for child care subsidies; some States have a second, higher threshold for families who already receive subsidies to allow for some income growth over time without impacting continued eligibility.

¹⁸ In addition, the Federal law requires States to prioritize services for specific populations including children of very low income families, children with special needs, and children

5.3.2. Benefit amount

CCDF childcare benefits amounts are set by the State and typically vary by type of provider and other factors. Families receiving CCDF subsidies are required to contribute to the price of care on a sliding scale (based at a minimum on family size and income), although states may waive co-payments for families at or below poverty.

Michigan's child care subsidies are based on a sliding scale related to gross monthly income and family size. The "entry" income ceiling for the initial subsidy eligibility per month ranges from USD 1,759 for a family of 2 to USD 2,665 for a family of 4. Co-payments range from USD 0 to USD 90 per child per bi-weekly pay period depending on income and subject to a maximum amount.¹⁹ Provider payment rate ceilings vary by age of child, type of care, and by locality. For centre-based care, the base rate ceiling is USD 4.00 per hour for infants/toddlers (under age 2 ½) and USD 2.75 per hour for preschool/school age (over age 2 ½).²⁰ Providers at higher quality ratings receive higher payments. Michigan waives co-payment fees for some families at or below the poverty level, specifically for families receiving TANF cash assistance and for families with children receiving protective services (not considered TaxBEN). Michigan pays its child care subsidies based on an hourly rate, which would be the same for full-day and part-time care. However, some States choose to base their payment rates on larger increments of time (i.e. daily). These payment rates will vary depending on whether the child was in full-time or part-time care. The amount of this variance is dependent on State policy.

5.3.3. Tax treatment

No States are known to treat CCDF or TANF assistance as taxable income source.

5.3.4. Interaction with other components of the tax-benefit system

When determining eligibility for CCDF, Supplemental Security Income (SSI) is fully counted in a little over half the States. TANF income is fully counted in a little under half of the States; it is either not counted or only counted under certain circumstances in the remainder. Most States do not count the value of energy assistance, housing assistance, the SNAP, or State EITC refunds.²¹

experiencing homelessness. States have flexibility as to how they offer priority to these populations, including by prioritizing enrolment, waiving co-payments, paying higher rates for access to higher-quality care, or using grants or contracts to reserve slots for priority populations. Many states offer priority for a child care subsidy eligibility to families receiving TANF cash assistance and to families transitioning off of TANF.

¹⁹ <https://dhhs.michigan.gov/OLMWEB/EX/RF/Public/RFT/270.pdf>, p.2. Families must have incomes below the "entry" ceiling to begin receiving subsidies. Then, their income can keep increasing past the "entry" amount, and they won't get kicked off until they hit the "exit" threshold. As TaxBEN assumes constant monthly earnings throughout the year and in order not to overestimate eligibility, only "entry" ceilings are considered in the model.

²⁰ <https://dhhs.michigan.gov/OLMWEB/EX/RF/Public/RFT/270.pdf>, p.3

²¹ [CCDF Policies Database Book of Tables: Key Cross-State Variations in CCDF Policies as of October 1, 2016, The Urban Institute.](#)

5.4. *Childcare allowance for children not using childcare centres*

None.

5.5. *Tax concessions for childcare expenditures (Child and Dependent Care Credit, CDCC)*

Variable name: [\[ActualCDCC_1; ActualCDCC_2\]](#)

The Child and Dependent Care Credit (CDCC) provides tax assistance to families who pay for child care in order to work in paid employment or look for work. The amount of credit is based on income, the number of dependents, and the amount of child care expenses. For example, families with adjusted gross income of less than or equal to USD 15 000 are eligible to receive a child care credit of 35 percent of qualifying child care expenses; however, note that because the credit is non-refundable, families that owe no (federal) tax will not receive the credit for which they are technically eligible. Families with higher income receive a lower credit, with the rate falling to 20 percent for individuals and couples with adjusted gross incomes above USD 43 000. In 2017, qualifying child care expenses were capped at USD 3 000 for one child and USD 6 000 for two or more children. Thus the maximum value of the credit was USD 1 050 for an individual or couple with one child and adjusted gross income below USD 15 000 and USD 2100 for a low-income family with two or more children. The credit is not refundable and so families that do not pay taxes (e.g., all families with married partners and incomes below USD 15 000) do not benefit from the credit at all.²²

6. In-work benefits

6.1. *Earned Income Tax Credit (EITC)*

Variable names: [\[IW; EIC_1; EIC_2; local_EIC_1; local_EIC_2\]](#)

The EITC is a refundable tax credit. Eligible for EITC are working families with children under 19 (or under 24 if full-time student, or any age if permanently or totally disabled) and childless working persons aged between 25 and 65 that meet certain income thresholds.

Workers with children: Low income workers with dependents are allowed a refundable (non-wastable) earned income credit. For taxpayers with one child, the credit is 34% of up to USD 10 180 of earned income in 2018. The credit phases down when income exceeds USD 18 660 (24 350 for married taxpayers) at a rate of 15.98% (21.06%) for taxpayers with one child (two and more children) and phases out when it reaches USD 40 320 (46 010 for married taxpayers). The earned income threshold and the phase-out threshold are indexed for inflation. For taxpayers with two children, the credit is 40% of up to USD 14 290 of earned income in 2018. The credit phases down when income exceeds USD 18 660 (24 350 for married taxpayers) and phases out when it reaches USD 45 802 (51 492 for married taxpayers). For taxpayers with three or more children the credit is 45% of up to USD 14 290 of earned income. The credit phases down when

²² See the “[Individual Income Tax Returns 2017](#)” Publication or the “[Child and Dependent Care Expenses Form 2241](#)” for more information.

income exceeds USD 18 660 (24 350 for married taxpayers) and phases out when it reaches USD 49 194 (54 884 for married taxpayers).

Workers without children: low income workers without children are also eligible for the EITC. In 2018 low income workers without children are permitted a non-wastable earned income credit of 7.65% of up to USD 6 780 of earned income. The credit phases down when income exceeds USD 8 490 (14 170 for married taxpayers) at a rate of 7.65% and phases out when income reaches USD 15 270 (20 950 for married taxpayers). This credit is available for taxpayers without children at least 25 years old and under 65 years old.

Michigan additionally applies a local EITC, see Section 8.2.

7. Social security contributions and payroll taxes

7.1. Employee's social security contributions

Variable names: `[SC; SOCSEC_p; SOCSEC_s]`

In 2018, the rate for employee contributions is 7.65% (6.2% for old age, survivors, and disability insurance, and 1.45% for old age hospital insurance). The 6.2% rate applies to earnings up to USD 128 400. Beginning in 1994, there is no limit on the amount of earnings subject to the 1.45% rate. There is an additional 0.9% tax on employee wages and salaries that exceed USD 200 000 (USD 250 000 for joint returns) as the additional hospital insurance tax on high-income taxpayers. The additional tax on wages and salaries is subject to withholding (but without regard to the earnings of the spouse) when wages from a particular job exceed USD 200 000 per year. These thresholds are not indexed for inflation.

7.2. Employer's social security contributions

Variable names: `[SSCR; SSCR_p; SSCR_s]`

Pensions: The rate for employers' contributions is 6.2% on earnings up to USD 128 400 and 1.45% of all earnings (without limit).

Unemployment: Employers are required by the federal government to pay unemployment tax of 6% on earnings up to USD 7 000. Taxes are also paid to various state-sponsored unemployment plans which may generally be credited against the required federal percentage. In 2018 the estimated average unemployment insurance tax rate in Michigan was 3.20% of the first USD 9 000 of wages. The model considers that the Federal government allows employers to take a credit for state unemployment taxes of up to 5.4%, resulting in a net Federal tax of 0.6% (6% minus 5.4%) on earnings up to USD 7 000.

7.3. Payroll taxes

See Section 7.1.

8. Taxes

8.1. Personal income tax

Variable names: `[IT; INCTAX_1; INCTAX_2]`

Taxation period is the calendar year.

Families in the United States are generally taxed in one of three ways:

- As married couples filing jointly on the combined income of both spouses;
- As married individuals filing separately and reporting actual income of each spouse;
- As heads of households (only unmarried or separated individuals with dependents).

All others, including dependent children with sufficient income, file as single individuals.

Married couples generally benefit from a more favourable schedule of tax rates for joint returns of spouses (see Section 8.1.2). Therefore, TaxBEN assumes that couples always file jointly.

8.1.1. Tax allowances and reliefs

Variable names: `[ALLOW_1; ALLOW_2; basic_1; basic_2; CH_CRED_TOT_1; CH_CRED_TOT_2]`

Basic allowances: In 2018 a married couple filing a joint tax return is entitled to a standard deduction of USD 24 000. The standard deduction is USD 18 000 for single heads of households (e.g. lone parents) and USD 12 000 for single individuals. This relief is indexed for inflation.^{23 24}

The additional **personal exemption** was abolished in 2018.

Earned Income Tax Credit (EITC): see Section 6.

Child credit: Taxpayers are permitted a tax credit for each qualifying child under the age of 17 (or 17- and 18-year-old children living at home; or 19- to 24-year-old children attending school full time). In 2018 the maximum credit is USD 2 000. The refundable (non-wastable) child credit is the lesser of 15% of earned income in excess of USD 2 500 and USD 1 400 per child. The refundable portion of the credit (USD 1 400 in 2018) is indexed for inflation and rounds down to the next lowest multiple of USD 100. For qualifying dependents other than qualifying children for whom a child tax credit was claimed, there is a USD 500 non-refundable credit (**other dependent tax credit**; not modelled in TaxBEN). Phase out of child tax credit and other dependent tax credit: The maximum credit is reduced for taxpayers with income in excess of certain thresholds. The total of the child tax credit and other dependent tax credit is reduced by USD 50 for each USD 1 000 by which modified aggregate gross income exceeds USD 400 000 for married

²³ More liberal standard deductions are available for taxpayers who are age 65 or older and taxpayers who are blind. Special rules apply to children who have sufficient income to pay tax and are also claimed as dependents by their parents.

²⁴ Alternatively to the standard basic allowance, it is possible to claim a basic non-standard relief which is the deduction of certain expenses to the extent that, when itemised, they exceed in aggregate the standard deduction. TaxBEN assumes that workers claim the standard allowance, so further details on the non-standard allowance are not provided.

taxpayers filing jointly (USD 200 000 for single and head of household – e.g. lone parent – taxpayers). These threshold amounts are indexed for inflation.

8.1.2. Definition of the tax base

Gross income minus the above tax exemptions.

8.1.3. Income tax schedule

Taxable Income Bracket (USD)*			Marginal Tax Rate (%)
Single Individual	Joint Return of Married Couple	Head of Household	
0 to 9 525	0 to 19 050	0 to 13 600	10
9 526 to 38 700	19 051 to 77 400	13 601 to 51 800	12
38 701 to 82 500	77 401 to 165 500	51 801 to 82 500	22
82 501 to 157 500	165 501 to 315 000	82 501 to 157 500	24
157 501 to 200 000	315 001 to 400 000	157 501 to 200 000	32
200 001 to 500 000	400 001 to 600 000	200 001 to 500 000	35
500 001 and over	600 001 and over	500 001 and over	37

* The taxable income brackets are indexed for inflation.

8.2. Local income taxes

Variable name: `[ST_LOC_TAX_1; ST_LOC_TAX_2]`

The District of Columbia and 41 of the 50 States impose some form of individual income tax. In addition, some local governments impose an individual income tax, although this is not generally the case. TaxBEN simulates the following local tax instruments as applied in Michigan:

- The state of Michigan permits a personal exemption of USD 4000 for the taxpayer and taxes income at the rate of 4.25 per cent.
- Michigan allows taxpayers who are eligible to claim the federal earned income tax credit (EITC; cp. Section 6) to claim a Michigan EITC. The Michigan EITC is a refundable (non-wastable) credit equal to 6 percent of the federal EITC.
- The city of Detroit permits a personal exemption of USD 600 and taxes income at the rate of 2.4 per cent.

Annex: Other benefits and direct taxes

This section provides a brief description of other cash benefits and taxes on employment incomes in the United States that are relevant for some members of the population below the statutory retirement age, but which are not included in the OECD tax-benefit model.

Supplemental Security Income (SSI)

SSI is a means-tested, federally administered income assistance programme which provides monthly cash payments in accordance with uniform, nation-wide eligibility requirements to needy aged, blind and disabled persons.

General Assistance programme (GA)

GA is a subsidy provided by local governments to needy individuals or families who do not qualify for the other major assistance programmes and to those whose benefits from other assistance programmes are insufficient to meet basic needs. Help may either be in cash or in kind, including such assistance as groceries and rent. Eligibility requirements and benefit amounts vary from State to State, and often within a State. Payments are usually at lower levels and of shorter duration than those provided by federally financed programmes. GA is administered and financed by State and local governments under their own guidelines.

Housing benefits

In the US, there are three primary rental assistance programs provide support for tenants subject to eligibility based on income: 1) public housing; 2) project-based Section 8 programs; 3) tenant-based Section 8 vouchers. Eligible income limits are based on a percentage of the local Area Median Income (AMI): no greater than 80 percent of AMI for both public housing and project-based Section 8 programs, and no greater than 50 percent of AMI for Section 8 vouchers, with adjustments for family size.

In practice, the three rental assistance programs serve tenants with much lower incomes - primarily below 30 percent of AMI. Local housing providers determine eligibility and admit tenants. Public housing authorities (PHAs) administer the public housing and tenant-based Section 8 programs, and private owners of assisted multifamily developments administer the project-based Section 8 program. Admission priorities are usually based on time on the local waiting list, but localities are allowed to establish local preferences (for example, for families transitioning to work from welfare, individuals with disabilities, and senior households).

The Department of Housing and Urban Development (HUD) calculates median incomes for the purpose of determining eligibility for federal rental assistance programs. For fiscal

year 2018, HUD estimated the national median income for a family of 4 at \$68,000,²⁵ with 80 percent of the median being \$54,400 and 30 percent being \$20,400. Using the example of the State of Michigan, the median income for a family of four was \$64,100 per year, with 80 percent being \$51,280 and 30 percent being \$19,230.

Tenants are required to pay 30 percent of their income for rent. Tenants in the Section 8 voucher programme are also allowed to pay more than that amount to access higher priced housing. There are often a series of deductions for rental payments for elderly and disabled persons, youth and families with specific expenses (e.g. child care).²⁶ In the public housing and tenant-based Section 8 programs, local PHAs are authorized to charge a “minimum rent” up to \$50 for families with very small incomes with hardship exceptions for some households.

In the public housing and project-based Section 8 programs, the benefit is the value of the housing provided. In the Section 8 voucher program, the subsidy makes up the difference between the tenant’s rent contribution and a locally determined Fair Market Rent. Again, using this Chapter’s example of the State of Michigan, the Fair Market Rent for the Detroit-Warren-Livonia metro area for a 2 bedroom apartment in 2018 was \$940 per month.²⁷

There is no statutory limit on duration of assistance. Families may lose assistance due to failure to comply with lease obligations, or through fraud or other criminal activity. Housing assistance is not taxable.

²⁵ Note that at the time of writing, the U.S. Department of Housing and Urban Development had not yet issued new income limits for FY 2018. The FY 2017 income limits remained in effect for its program operations. See “[FY 2017 HUD Income Limits Briefing Materials](#)” page 38 (“Attachment 6”).

²⁶ Several types of income deductions reduce the required tenant rent contributions for families with a head of household who is elderly or disabled, medical expenses that exceed 3 percent of income per year, “reasonable” child care expenses, minors, and adults who are full-time students (or attending vocational training, disabled family members). In addition, child support, spousal support and income earned by minors are not counted toward income (with some limits) for purposes of calculating rent.

²⁷ More information on calculation of Fair Market Rents can be found at <https://www.huduser.gov/portal/datasets/fmr.html>.